# Yan Yang [Google Scholar](#) · [GitHub](#) · [LinkedIn](#)

Machine learning engineer with industry experience in deep learning, generative models, and ML embedding visualization system design. Research background in machine learning, PDE learning, user motivation/affection modeling, information retrieval, and computer vision. M.Sc. in Computer Science (Georgetown University).

## SKILLS

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|---|---|
| **ML / AI** | PyTorch, GANs, CNNs, backpropagation, PDE learning, PINN, embedding visualization, data augmentation |
| **Languages** | Python, C++, JavaScript/TypeScript, GoLang, SQL |
| **Systems** | System design, API architecture, database schemas, caching, serialization, Kubernetes, Redis |
| **Research** | Reaction-diffusion systems, NLP, information retrieval, PDE modeling, fMRI processing/model, EEG |
| **Tools** | Git, Linux, NumPy, React, Node.js, AngularJS, FreeSurfer |

## EXPERIENCE

**Machine Learning R&D Engineer** | Deepcell Inc. · Mountain View, CA *2022 – 2023*

- Led system design of the remodeled neural network embedding visualization system, integrated into the Cluster-Based Sorting product for real-time data exploration during imaging+sorting runs.
- Architecture decisions spanned data storage, database schemas, API specs, serialization/deserialization, caching, parallelization, and incremental development planning.

**Research Fellow** | KAUST · Thuwal, Saudi Arabia *2023 – 2024*

- Investigated data-driven methods for learning PDE parameters in reaction-diffusion systems; the underlying mathematics connects to neuroscience modeling, and in this project was applied to automate synthesis of diseased leaf images for agricultural classification models.
- Proposed approaches to Arabic speech recognition, including acoustic source separation and spectrogram-based data augmentation.

**Independent Researcher** | Self-directed *Oct 2025 – Present*

- Continued KAUST-era research on reaction-diffusion system parameter learning; presented solo-author paper at AI4Physics workshop @ICML'26; built a code repository with visualization and logging tools for dynamical system loss landscape analysis.
- Experimented with direct backpropagation through unrolled Gray-Scott time steps, adaptive learning rates, and alternative loss functions; evaluated PINN and surrogate model approaches.
- Published conceptual discussion preprints on user motivation/affection modeling.

**R&D Intern** | Deepcell Inc. · Mountain View, CA *2021 – 2022*

- Trained GAN models to supplement data for cell image classifiers.
- Led development of a CNN embedding layer visualization tool used company-wide by data and bioinformatics scientists for clustering and correlation analysis.

**Member of Technical Staff** | Pure Storage · Bellevue, WA *2020 – 2021*

- Implemented volume snapshot import/restore from company disks into Kubernetes.
- Bug fixes and enhancements: multipath device issues, Python 3 migration of bash scripts; led GitHub release process for Pure Service Orchestrator v6.0.3.

**Research Assistant** | Georgetown University – InfoSense Lab · Washington, DC *2018 – 2019*

- Translated reversal theory into statistical frameworks to model search engine user behavior and improve struggle detection; work published in Discover Computing (Springer Nature, 2024).
- Revamped VR interface for behavioral data collection; provided web support for SIGIR '18 conference demonstration, acknowledged as authorship-worthy by corresponding author.

*Earlier experience includes SDE roles at ColorfulClouds Tech, DataMesh, etc. (2013-2018). Full history available on request.*

## EDUCATION

**Ph.D. studies, Computer Science** | University of Nevada, Reno · Reno, NV *2021 – 2022*

**M.Sc., Computer Science** | Georgetown University · Washington, DC *2017 – 2019*

**B.Eng.** | Southeast University · Nanjing, China *2002 – 2006*

## SELECTED PUBLICATIONS & OUTPUT

- Yang (2026). Loss Landscape Diagnosis for Gradient-Based Inversion: Disentangling PINN Components. AI4Physics@ICML'26
- Jiyun et al. (2024). Improving Searcher Struggle Detection via the Reversal Theory. Discover Computing 27(51), Springer Nature
- Yang et al. (2022). Two-Step Data Augmentation for Masked Face Detection and Recognition. IMPROVE Conference, SciTePress

## PEER REVIEW

IEEE Transactions on Affective Computing 2025 · IEEE Transactions on Neural Networks and Learning Systems 2024